

OC Core 1.3 / Biology Domain Packet

Status Snapshot

- Current state: PACKETIZED_PENDING_QUANTITATIVE_VALIDATION
- Target state: EMPIRICAL_HARD_CLOSED
- Quantitative pass result: FAIL_CLOSED
- Next required action: EXECUTE_BIOLOGY_NUMERICAL_REPLAY_PACKET
- Replay status: REPLAY_HARNESS_READY_PENDING_EXECUTION

Theorem-to-Observable Map

- K4/K5/K6/K7 biological organization claims are restricted to bounded state-transition and response-signature lanes.
- The packet currently targets measurable expression and cell-state trajectories rather than unrestricted biological universalization.
- Open biological assay repositories are the first-line evidence route; if insufficient, the program escalates to institute-run measurement.

Benchmark Dataset Manifest

- {'authority_ref': 'NCBIGEO', 'benchmark_id': 'BIO_BENCH_001', 'dataset_url': 'https://www.ncbi.nlm.nih.gov/geo/', 'held_out_policy': 'reserve one time-series family for held-out replay', 'observable_name': 'expression_signature_residual'}
- {'authority_ref': 'NCBIGEO', 'benchmark_id': 'BIO_BENCH_002', 'dataset_url': 'https://www.ncbi.nlm.nih.gov/geo/', 'held_out_policy': 'reserve one cell-state transition family', 'observable_name': 'state_transition_order_error'}
- {'authority_ref': 'official open biological assay repositories', 'benchmark_id': 'BIO_BENCH_003', 'dataset_url': 'https://www.ncbi.nlm.nih.gov/geo/', 'held_out_policy': 'reserve one response-signature family', 'observable_name': 'response_onset_residual'}

Measurable Outputs

- expression_signature_residual
- state_transition_order_error
- response_onset_residual

Measurement Schema

Choose one bounded biological lane and evaluate predicted state transitions or response signatures against open assay data.

Acceptance Criterion

The promoted biological packet reproduces the declared measurable signatures on held-out datasets without widening the claim boundary.

Falsifier

If the bounded signature family fails on held-out datasets, the biological packet remains frontier-bound.

Replay Harness

- Replay command: `logion/k7/spe/orchestrator/science/run_oc_core_domain_hard_closure_replay_v1.py`
- Replay program status: PROGRAM_DEFINED_PENDING_EXECUTION

Official / Open Source Routes

- U.S. National Center for Biotechnology Information (OFFICIAL_OPEN_DATA_HUB): <https://www.ncbi.nlm.nih.gov/geo/>

Blocking IDs

- BIOLOGY__NUMERICAL_PACKET_PASS_REQUIRED
- BIOLOGY__REPLAY_PASS_REQUIRED